# Tianzhi Li

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## Basic Information

I am an incoming **Shun Hing Postdoctoral Fellow** at the Chinese University of Hong Kong (CUHK), working with Prof. Zhongyu Li. I received my Ph.D. in Mechanics & Control from Peking University (PKU) in June 2026, working with Prof. Jinzhi Wang and Prof. Zhisheng Duan. From Dec-2024 to Dec-2025, I was a visiting PhD at Nanyang Technological University (NTU), Singapore, advised by Prof. François Gay-Balmaz. Prior to that, I received my B.Sc. in Mathematics from Beijing Institute of Technology (BIT) in 2021 under the supervision of Prof. Donghua Shi.

My research interest lies at the intersection of **Geometric Mechanics**, **Geometric Robot Learning**, and **Geometric Embodied AI**. I am broadly interested in leveraging physical principles and differential geometry to develop computationally efficient algorithms for **control**, **estimation**, and **learning** with sound theoretical guarantees.

## Appointments

- **Postdoctoral Fellow**
  Department of Mechanical and Automation Engineering
  The Chinese University of Hong Kong

## Education

- **Peking University** Sep 2021 - Jul 2026
  PhD (Dynamical Systems and Control)
- **Nanyang Technological University**, Singapore Dec 2024 - Dec 2025
  Visiting PhD Student (Geometric Mechanics)
- **Beijing Institute of Technology** Sep 2017 - Jun 2021
  Bachelor (Mathematics)

## Teaching Experience

- Teaching Assistant, "Ordinary Differential Equations", Peking University Fall 2023
  Undergraduate-level course, covering more than 150 students
  Recieved **Outstanding Teaching Assistant Award**
- Teaching Assistant, "Linear Algebra", Peking University Spring 2023
  Undergraduate-level course, covering more than 150 students
- Teaching Assistant, "Linear Algebra", Peking University Fall 2022
  Undergraduate-level course, covering more than 150 students
  Recieved **Outstanding Teaching Assistant Award**
- Teaching Assistant, "Analytic Mechanics", Peking University Spring 2022
  Undergraduate-level course, covering more than 100 students
- Teaching Assistant, "Ordinary Differential Equations", Peking University Fall 2021
  Undergraduate-level course, covering more than 150 students

## Publications

### Preprints / Under Review

[P1] Y. Lin, Z. Duan, **T. Li**, N. Bai, and Z. Sun, Characterizing Robustness in Nonlinear Optimal Control: From Stability to Optimality, under review.

[P2] M. Du, F. Gay-Balmaz, **T. Li**, and D. Shi, Stochastic Geometrically Exact Beam: Continuous-Time Geometric Principles and Variational Discretizations, to be submitted.

**Published Papers**

[1] **T. Li** and J. Wang, Variational Unscented Kalman Filter on Matrix Lie Groups, **Automatica** (Regular Paper), 172: 111995, 2025. [Paper Link]

[2] **T. Li**, R. Fu, and J. Wang, Reduced Dynamics and Geometric Optimal Control of Nonequilibrium Thermodynamics: Gaussian Case, **Automatica** (Regular Paper), 164: 111626, 2024. [Paper Link]

[3] **T. Li** and J. Wang, Physics-Informed Gaussian Process Learning on Lie Groups, **Journal of Guidance, Control, and Dynamics**, 48 (11), pp. 2654-2662, 2025. [Paper Link]

[4] **T. Li**, J. Wang, and Z. Duan, Structure-Preserving Unscented Kalman Filter for Planar Mobile Robots, **IEEE Control Systems Letters**, vol. 9, pp. 2157-2162, 2025. [Paper Link]

[5] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic Systems **Part II**: Stochastic Nonholonomic Integrator. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), Saint-Malo, France, vol. 16034, pp. 225-233. Springer, 2026. [Paper Link]

[6] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic Systems **Part I**: Continuous-Time Formulation. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), France, vol. 16034, pp. 204-213. Springer, 2026. [Paper Link]

[7] **T. Li** and J. Wang, A Structure-Preserving Learning Scheme on SO(3), 2024 43rd IEEE Chinese Control Conference (**CCC'24**), Kunming, China, 2024, pp. 5149-5152. [Paper Link]

[8] **T. Li** and J. Wang, Multisymplectic Unscented Kalman Filter for Geometrically Exact Beams. In: Nielsen, F., Barbaresco, F. (eds) International Conference on Geometric Science of Information (**GSI'23**). Lecture Notes in Computer Science, Saint-Malo, France, vol. 14072, pp. 60-68, Springer Verlag. [Paper Link]

[9] **T. Li** and J. Wang, A Physics-Informed Gaussian Process Regression Algorithm for The Dynamics of The Planar Pendulum, 2023 42nd IEEE Chinese Control Conference (**CCC'23**), Tianjin, China, pp. 5163-5167, 2023. [Paper Link]

[10] **T. Li** and J. Wang, A Statistical Dynamical Algorithm for Gaussian Multi-Agent Systems Under Hamel's Formalism, 34th IEEE Chinese Control and Decision Conference (**CCDC'22**), Hefei, China, pp. 1344-1349, 2022. [Paper Link]

## Honors and Awards

- **Outstanding Graduate of Beijing** — Jun-2026
  Issued by Beijing Municipal Commission of Education
- **Outstanding Graduate of Peking University** — Jun-2026
  Issued by Peking University
- **National Scholarship** (top 1%) — Sep-2025
  Issued by Ministry of Education
- **Merit Student of Beijing** — Dec-2025
  Issued by Beijing Municipal Commission of Education
- **Presidential Doctoral Scholarship** — Jun-2025
  (The most prestigious scholarship issued by Peking University)
- **Merit Student Award** (top 1%) — Sep-2025
  Issued by Peking University
- **Excellent Academic Research Award** (top 1%) — Sep-2025
  (only 2 awardees at SAMR, Peking University)
- **Dean's Doctoral Scholarship - First Prize** — Oct-2024
  Issued by College of Engineering, Peking University

- **Yuehua Luo Scholarship** Nov-2024
  Issued by Peking University
- **Excellent Teaching Assistant Award** (× 2) Mar-2024 & Apr-2023
  Issued by College of Engineering, Peking University

## Conference Talks & Invited Talks

- Variational Principle for Stochastic Nonholonomic Systems Part II: Stochastic Nonholonomic Integrator
  The International Conference on Geometric Science of Information (GSI)
  Saint-Malo, France, Oct-2025
- Variational Principle for Stochastic Nonholonomic Systems Part I: Continuous-Time Formulation
  The International Conference on Geometric Science of Information (GSI)
  Saint-Malo, France, Oct-2025
- Geometric Optimal Control of Gaussian Distributions (Invited by Profs. Qiang Tian and Ju Chen)
  Workshop on Multibody System Dynamics
  School of Aerospace Engineering, Beijing Institute of Technology, Apr-2024.
- Variational Unscented Kalman Filter on Matrix Lie Groups (Invited by Prof. Donghua Shi)
  Geometric Mechanics Control Seminar
  School of Mathematics and Statistics, Beijing Institute of Technology, Jul-2024.

## Academic Service

**Journal Reviewer**: Automatica, International Journal of Dynamics and Control.

**Conference Reviewer**: IEEE Conference on Decision and Control (CDC), IFAC Workshop on Lagrangian and Hamiltonian Methods for Nonlinear Control (LHMNC), International Symposium on Mathematical Theory of Networks and Systems (MTNS).

## References

**Professor Jinzhi Wang**
Position: Professor of Dynamics and Control
Affiliation: Peking University
E-mail: jinzhiw@pku.edu.cn

**Professor Donghua Shi**
Position: Professor of Mathematics
Director of Beijing Key Laboratory on MCAACI
Affiliation: Beijing Institute of Technology
E-mail: dshi@bit.edu.cn

**Professor François Gay-Balmaz**
Position: Associate Professor of Mathematics
Affiliation: Nanyang Technological University
E-mail: francois.gb@ntu.edu.sg

**Professor Zhisheng Duan**
Position: Professor of Dynamics and Control
Deputy Director of State Key Laboratory for Turbulence & Complex Systems
Affiliation: Peking University
E-mail: duanzs@pku.edu.cn